Supplementary Materials

Supplementary Appendix A. The Proof of *Lemma 1*. This appendix provides a detailed proof of *Lemma 1* in Section III-B2 of the manuscript.

According to the *Frenet–Serret formulas*, there is

$$\begin{aligned}\mathbf{t}' &= \kappa\mathbf{n},\\ \mathbf{n}' &= -\kappa\mathbf{t} + \tau\mathbf{b},\\ \mathbf{b}' &= -\tau\mathbf{n}.\end{aligned}$$

According to $\boldsymbol{\gamma}' = \mathbf{t}$, there is

$$\frac{\partial \mathbf{X}}{\partial l} = \mathbf{t} + d_n(-\kappa\mathbf{t} + \tau\mathbf{b}) - \tau d_b\mathbf{n} = (1-\kappa d_n)\mathbf{t} - \tau d_b\mathbf{n} + \tau d_n\mathbf{b}.$$

Differentiating $\mathbf{X}(\mathbf{q})$ with respect to d_n and d_b gives $\partial\mathbf{X}/\partial d_n = \mathbf{n}$ and $\partial\mathbf{X}/\partial d_b = \mathbf{b}$. Therefore, Eq. (20) of the manuscript is proved. Then, according to Eq. (20) of the manuscript, there is

$$\begin{aligned}\det\Big(\frac{\partial \mathbf{X}}{\partial \mathbf{q}}\Big) &= \det\big[(1-\kappa d_n)\mathbf{t} - \tau d_b\mathbf{n} + \tau d_n\mathbf{b},\ \mathbf{n},\ \mathbf{b}\big]\\ &= (1-\kappa d_n)\det[\mathbf{t},\mathbf{n},\mathbf{b}] - \tau d_b\det[\mathbf{n},\mathbf{n},\mathbf{b}] + \tau d_n\det[\mathbf{b},\mathbf{n},\mathbf{b}]\\ &= (1-\kappa d_n)\det[\mathbf{t},\mathbf{n},\mathbf{b}]\\ &= (1-\kappa d_n)\,\mathbf{t}^{\mathrm{T}}(\mathbf{n}\times\mathbf{b}) = 1-\kappa d_n,\end{aligned}$$

where $\mathbf{n}\times\mathbf{b} = \mathbf{n}\times(\mathbf{t}\times\mathbf{n}) = \mathbf{t}$. Since $\kappa \geq 0$, $|d_n| \leq \lambda$, and $\kappa\lambda < 1$, there is $\kappa d_n < 1$, hence $J = 1-\kappa d_n > 0$. Next, substituting Eq. (20) of the manuscript into Eq. (19) of the manuscript, there is

$$\mathbf{v}_{\mathrm{d}} = \dot{l}\frac{\partial \mathbf{X}}{\partial l} + \dot{d}_n\mathbf{n} + \dot{d}_b\mathbf{b} = J\dot{l}\,\mathbf{t} + (\dot{d}_n - \tau d_b\dot{l})\mathbf{n} + (\dot{d}_b + \tau d_n\dot{l})\mathbf{b}. \tag{S1}$$

Comparing the components of Eq. (S1) with those of $\mathbf{v}_{\mathrm{d}}$ in Eq. (12) of the manuscript, there is

$$J\dot{l} = v_{\mathrm{d},l}, \quad \dot{d}_n - \tau d_b\dot{l} = v_{\mathrm{d},d_n}, \quad \dot{d}_b + \tau d_n\dot{l} = v_{\mathrm{d},d_b},$$

which gives Eq. (21) of the manuscript. In 2D, $\tau = 0$, the d_b and $\mathbf{b}$ terms are absent, so the same steps give $\partial\mathbf{X}/\partial l = (1-\kappa d_n)\mathbf{t}$, and $\dot{d}_n = v_{\mathrm{d},d_n}$.

This completes the proof.

Supplementary Appendix B. The Proof of *Proposition 1*. This appendix provides a detailed proof of *Proposition 1* in Section III-B2 of the manuscript.

In the manuscript, the transverse fluxes F_n and F_b are defined by

$$\begin{aligned}F_n &\triangleq J\rho_{\mathrm{d}}\dot{d}_n = J\rho_{\mathrm{d}}v_{\mathrm{d},d_n} + \tau d_b\rho_{\mathrm{d}}v_{\mathrm{d},l},\\ F_b &\triangleq J\rho_{\mathrm{d}}\dot{d}_b = J\rho_{\mathrm{d}}v_{\mathrm{d},d_b} - \tau d_n\rho_{\mathrm{d}}v_{\mathrm{d},l},\end{aligned} \tag{S2}$$

where the second equalities follow from Eq. (21) of the manuscript, which also gives $J\rho_{\mathrm{d}}\dot{l} = \rho_{\mathrm{d}}v_{\mathrm{d},l}$.

(i) For symmetric cross sections, $\mathcal{A}(l)$ is the disc $\|\boldsymbol{\zeta}\| \leq 1$ of radius $\lambda(l)$, whose boundary $\partial\mathcal{A}(l)$ has the outward unit normal $\boldsymbol{\zeta}$ and expands with l at the rate λ'. Integrating Eq. (26) of the manuscript over $\mathcal{A}(l)$, there is

$$\int_{\mathcal{A}(l)}\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}\,\mathrm{d}A + \int_{\mathcal{A}(l)}\Big(\frac{\partial F_n}{\partial d_n} + \frac{\partial F_b}{\partial d_b}\Big)\mathrm{d}A = 0. \tag{S3}$$

According to the *Divergence Theorem* in the (d_n, d_b) plane and Eq. (28) of the manuscript, there is

$$\int_{\mathcal{A}(l)}\Big(\frac{\partial F_n}{\partial d_n} + \frac{\partial F_b}{\partial d_b}\Big)\mathrm{d}A = \oint_{\partial\mathcal{A}(l)}\boldsymbol{\zeta}^{\mathrm{T}}\mathbf{F}_{\perp}\,\mathrm{d}s = \oint_{\partial\mathcal{A}(l)}\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\,\mathrm{d}s, \tag{S4}$$

where $\mathrm{d}s$ is the length element of $\partial\mathcal{A}(l)$. Since the integration domain $\mathcal{A}(l)$ depends on l and its boundary moves outward at the rate λ', the *Leibniz integral rule* and Eq. (29) of the manuscript give

$$\frac{\mathrm{d}Q}{\mathrm{d}l} = \frac{\mathrm{d}}{\mathrm{d}l}\int_{\mathcal{A}(l)}\rho_{\mathrm{d}}v_{\mathrm{d},l}\,\mathrm{d}A = \int_{\mathcal{A}(l)}\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}\,\mathrm{d}A + \oint_{\partial\mathcal{A}(l)}\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\,\mathrm{d}s. \tag{S5}$$

Substituting Eq. (S4) into Eq. (S3) and comparing with Eq. (S5), there is $\mathrm{d}Q/\mathrm{d}l = 0$, thus $Q(l)$ is a constant Q_0. According to Eq. (29) of the manuscript, this is Eq. (30) of the manuscript.

(ii) Suppose that Eq. (30) of the manuscript holds. For each l, let ϕ be a solution of the Neumann problem

$$\begin{aligned}&\frac{\partial^2\phi}{\partial d_n^2}+\frac{\partial^2\phi}{\partial d_b^2}=-\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}\quad\text{in }\mathcal{A}(l),\\&\frac{1}{\lambda}\Big(d_n\frac{\partial\phi}{\partial d_n}+d_b\frac{\partial\phi}{\partial d_b}\Big)=\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\quad\text{on }\partial\mathcal{A}(l).\end{aligned}\tag{S6}$$

Integrating the first equation of Eq. (S6) over $\mathcal{A}(l)$ and applying the divergence theorem with the second equation, a solution exists if and only if

$$\int_{\mathcal{A}(l)}\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}\,\mathrm{d}A+\oint_{\partial\mathcal{A}(l)}\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\,\mathrm{d}s=0.\tag{S7}$$

According to Eq. (S5), the left side of Eq. (S7) equals $\mathrm{d}Q/\mathrm{d}l$, which is zero because $Q=v_{\mathrm{d},l}M=Q_0$ by the condition in Eq. (30) of the manuscript. Hence, ϕ exists.

Next, let $F_n=\partial\phi/\partial d_n$ and $F_b=\partial\phi/\partial d_b$. Then, according to Eq. (S6),

$$\begin{aligned}&\frac{\partial F_n}{\partial d_n}+\frac{\partial F_b}{\partial d_b}=-\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}\quad\text{in }\mathcal{A}(l),\\&\boldsymbol{\zeta}^{\mathrm{T}}\mathbf{F}_\perp=\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\quad\text{on }\partial\mathcal{A}(l),\end{aligned}$$

which are Eqs. (26) and (28) of the manuscript. Since $J>0$ by *Lemma 1* and $\rho_{\mathrm{d}}>0$, Eq. (S2) gives the transverse velocity components

$$v_{\mathrm{d},d_n}=\frac{F_n-\tau d_b\rho_{\mathrm{d}}v_{\mathrm{d},l}}{J\rho_{\mathrm{d}}},\qquad v_{\mathrm{d},d_b}=\frac{F_b+\tau d_n\rho_{\mathrm{d}}v_{\mathrm{d},l}}{J\rho_{\mathrm{d}}},$$

with which $\mathbf{v}_{\mathrm{d}}$ satisfies Eqs. (26) and (28) of the manuscript at every point of the tube.

This completes the proof.

Supplementary Appendix C. The Proof of *Lemma 2*. This appendix provides a detailed proof of *Lemma 2* in Section III-B2 of the manuscript.

According to Eqs. (11) and (15) of the manuscript, and since w has unit mean over $\mathcal{D}$, $M=\int_{\mathcal{A}(l)}\rho_{\mathrm{d},l}\,w(\boldsymbol{\zeta})\,\mathrm{d}A=\rho_{\mathrm{d},l}S$. Hence, according to Eq. (31) of the manuscript, there is

$$\rho_{\mathrm{d}}v_{\mathrm{d},l}=\rho_{\mathrm{d},l}w(\boldsymbol{\zeta})\frac{Q_0}{\rho_{\mathrm{d},l}S}=\frac{Q_0\,w(\boldsymbol{\zeta})}{S}.\tag{S8}$$

According to Eqs. (S2) and (S8), Eq. (32) of the manuscript, and $J\dot{l}=v_{\mathrm{d},l}$ in Eq. (21) of the manuscript, the transverse flux is

$$\mathbf{F}_\perp=J\rho_{\mathrm{d}}[\dot{d}_n,\dot{d}_b]^{\mathrm{T}}=\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\boldsymbol{\zeta}=\frac{Q_0\lambda'}{\lambda S}\,w(\boldsymbol{\zeta})\,[d_n,d_b]^{\mathrm{T}}.\tag{S9}$$

First, it is proved that Eq. (32) of the manuscript satisfies Eq. (26) of the manuscript. In 3D, according to Eq. (14) of the manuscript and $\|\boldsymbol{\zeta}\|^2=(d_n^2+d_b^2)/\lambda^2$, there is

$$\begin{aligned}w&=1+\mu_w-\frac{2\mu_w(d_n^2+d_b^2)}{\lambda^2},\\\frac{\partial w}{\partial l}&=-2\mu_w(d_n^2+d_b^2)\frac{\partial\lambda^{-2}}{\partial l}=-2\mu_w(d_n^2+d_b^2)\Big(-\frac{2\lambda'}{\lambda^3}\Big)=\frac{4\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda},\\\frac{\partial w}{\partial d_n}&=-\frac{4\mu_w d_n}{\lambda^2},\\\frac{\partial w}{\partial d_b}&=-\frac{4\mu_w d_b}{\lambda^2}.\end{aligned}\tag{S10}$$

Differentiating Eq. (S8) with respect to l by the quotient rule and substituting Eq. (S10), there is

$$\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}=\frac{Q_0}{S}\frac{\partial w}{\partial l}-\frac{Q_0\,w\,S'}{S^2}=\frac{Q_0}{S}\Big(\frac{4\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}-\frac{S'}{S}\,w\Big).\tag{S11}$$

According to Eq. (S9), $F_n=Q_0\lambda' w\,d_n/(\lambda S)$ and $F_b=Q_0\lambda' w\,d_b/(\lambda S)$, where $Q_0\lambda'/(\lambda S)$ depends only on l. Differentiating F_n with respect to d_n and F_b with respect to d_b by the product rule and substituting Eq. (S10), there is

$$\begin{aligned}\frac{\partial F_n}{\partial d_n}+\frac{\partial F_b}{\partial d_b}&=\frac{Q_0\lambda'}{\lambda S}\Big(w+d_n\frac{\partial w}{\partial d_n}\Big)+\frac{Q_0\lambda'}{\lambda S}\Big(w+d_b\frac{\partial w}{\partial d_b}\Big)\\&=\frac{Q_0\lambda'}{\lambda S}\Big(2w-\frac{4\mu_w d_n^2}{\lambda^2}-\frac{4\mu_w d_b^2}{\lambda^2}\Big)=\frac{Q_0}{S}\Big(\frac{2\lambda'}{\lambda}\,w-\frac{4\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}\Big).\end{aligned}\tag{S12}$$

Since $S=\pi\lambda^2$ gives $S'/S=2\lambda'/\lambda$, adding Eqs. (S11) and (S12), there is

$$\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}+\frac{\partial F_n}{\partial d_n}+\frac{\partial F_b}{\partial d_b}=\frac{Q_0\,w}{S}\Big(\frac{2\lambda'}{\lambda}-\frac{S'}{S}\Big)=0,$$

which is Eq. (26) of the manuscript. In 2D, F_b is absent and $\|\boldsymbol{\zeta}\|^2=d_n^2/\lambda^2$. According to Eq. (14) of the manuscript, there is

$$\begin{aligned}
w&=1+\mu_w-\frac{3\mu_w d_n^2}{\lambda^2},\\
\frac{\partial w}{\partial l}&=-3\mu_w d_n^2\frac{\partial\lambda^{-2}}{\partial l}=-3\mu_w d_n^2\Big(-\frac{2\lambda'}{\lambda^3}\Big)=\frac{6\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda},\\
\frac{\partial w}{\partial d_n}&=-\frac{6\mu_w d_n}{\lambda^2},
\end{aligned}\tag{S13}$$

Differentiating Eq. (S8) with respect to l and substituting Eq. (S13), there is

$$\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}=\frac{Q_0}{S}\frac{\partial w}{\partial l}-\frac{Q_0\,w\,S'}{S^2}=\frac{Q_0}{S}\Big(\frac{6\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}-\frac{S'}{S}\,w\Big).\tag{S14}$$

Differentiating F_n with respect to d_n and substituting Eq. (S13), there is

$$\frac{\partial F_n}{\partial d_n}=\frac{Q_0\lambda'}{\lambda S}\Big(w+d_n\frac{\partial w}{\partial d_n}\Big)=\frac{Q_0\lambda'}{\lambda S}\Big(w-\frac{6\mu_w d_n^2}{\lambda^2}\Big)=\frac{Q_0}{S}\Big(\frac{\lambda'}{\lambda}\,w-\frac{6\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}\Big).\tag{S15}$$

Since $S=2\lambda$ gives $S'/S=\lambda'/\lambda$, adding Eqs. (S14) and (S15), there is

$$\frac{\partial(\rho_{\mathrm{d}}v_{\mathrm{d},l})}{\partial l}+\frac{\partial F_n}{\partial d_n}=\frac{Q_0\,w}{S}\Big(\frac{\lambda'}{\lambda}-\frac{S'}{S}\Big)=0,$$

which is Eq. (26) of the manuscript.

Next, it is proved that Eq. (32) of the manuscript satisfies Eq. (28) of the manuscript. On $\partial\mathcal{A}(l)$, $\|\boldsymbol{\zeta}\|=1$. According to Eq. (S9), there is $\boldsymbol{\zeta}^{\mathrm{T}}\mathbf{F}_\perp=\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'\|\boldsymbol{\zeta}\|^2=\rho_{\mathrm{d}}v_{\mathrm{d},l}\lambda'$, which is Eq. (28) of the manuscript.

Finally, the physical components of Eq. (32) of the manuscript are derived. According to Eq. (32) of the manuscript, $\boldsymbol{\zeta}=[d_n,d_b]^{\mathrm{T}}/\lambda$, and $\dot{l}=v_{\mathrm{d},l}/J$ in Eq. (21) of the manuscript, there is

$$\begin{aligned}
\dot{d}_n&=\dot{l}\lambda'\frac{d_n}{\lambda}=\frac{v_{\mathrm{d},l}\lambda' d_n}{\lambda J},\\
\dot{d}_b&=\dot{l}\lambda'\frac{d_b}{\lambda}=\frac{v_{\mathrm{d},l}\lambda' d_b}{\lambda J}.
\end{aligned}\tag{S16}$$

According to Eq. (21) of the manuscript and Eq. (S16), there is

$$\begin{aligned}
v_{\mathrm{d},d_n}&=\dot{d}_n-\frac{\tau d_b v_{\mathrm{d},l}}{J}=\frac{v_{\mathrm{d},l}\lambda' d_n}{\lambda J}-\frac{\tau\lambda d_b v_{\mathrm{d},l}}{\lambda J}=\frac{v_{\mathrm{d},l}\big(\lambda' d_n-\tau\lambda d_b\big)}{\lambda J},\\
v_{\mathrm{d},d_b}&=\dot{d}_b+\frac{\tau d_n v_{\mathrm{d},l}}{J}=\frac{v_{\mathrm{d},l}\lambda' d_b}{\lambda J}+\frac{\tau\lambda d_n v_{\mathrm{d},l}}{\lambda J}=\frac{v_{\mathrm{d},l}\big(\lambda' d_b+\tau\lambda d_n\big)}{\lambda J},
\end{aligned}$$

which is Eq. (34) of the manuscript. In 2D, $\tau=0$ and d_b is absent, so Eq. (21) of the manuscript gives $v_{\mathrm{d},d_n}=\dot{d}_n$, and Eq. (S16) gives $v_{\mathrm{d},d_n}=v_{\mathrm{d},l}\lambda' d_n/(\lambda J)$, which is Eq. (33) of the manuscript.

This completes the proof.

Supplementary Appendix D. The Proof of *Proposition 2*. This appendix provides a detailed proof of *Proposition 2* in Section III-B3 of the manuscript.

First, it is proved that the planned longitudinal velocity has a positive lower bound. Since the swarm is confined by the tube boundaries in the d_n and d_b directions, there is no swarm inflow or outflow across the tube boundaries, which is the no-flux condition in Eq. (28) of the manuscript. According to Eqs. (11), (14), and (15) of the manuscript, $w(\boldsymbol{\zeta})$ has unit mean over the cross section, and there is

$$M(l)=\int_{\mathcal{A}(l)}\rho_{\mathrm{d}}\,\mathrm{d}A=\rho_{\mathrm{d},l}(l)\int_{\mathcal{A}(l)}w(\boldsymbol{\zeta})\,\mathrm{d}A=\rho_{\mathrm{d},l}(l)S(l).\tag{S17}$$

For any feasible solution of Eq. (35) of the manuscript, $\rho_{\mathrm{d}}\le\rho_{\max}$ holds in the tube. Hence,

$$M(l)=\int_{\mathcal{A}(l)}\rho_{\mathrm{d}}\,\mathrm{d}A\le\rho_{\max}S(l)\le\rho_{\max}S_{\max}.\tag{S18}$$

Substituting Eq. (S18) into Eq. (30) of the manuscript, there is

$$v_{\mathrm{d},l}(l) = \frac{Q_0}{M(l)} \geq \frac{Q_0}{\rho_{\max} S_{\max}} > 0,$$

where $Q_0 > 0$ according to Eq. (35) of the manuscript. Hence, the planned longitudinal velocity has a positive lower bound. This bound holds for any feasible solution of Eq. (35) of the manuscript and does not require the planned density to equal the desired density.

Furthermore, when the planned density equals the desired density in Eq. (13) of the manuscript, $\rho_{\mathrm{d},l} = \rho_{\mathrm{g},l}$. Substituting Eq. (S17) into Eq. (30) of the manuscript, there is

$$\rho_{\mathrm{g},l}(l) v_{\mathrm{d},l}(l) S(l) = Q_0. \tag{S19}$$

Substituting the desired density in Eq. (13) of the manuscript,

$$\rho_{\mathrm{g},l}(l) = \rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha},$$

into the above Eq. (S19) yields

$$v_{\mathrm{d},l}(l) = \frac{Q_0}{\rho_{\mathrm{g},l}(l) S(l)} = \frac{Q_0}{\rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha} S(l)} = \frac{Q_0}{\rho_0 S_0^{\alpha}} S(l)^{\alpha - 1}.$$

Let

$$C_1 = \frac{Q_0}{\rho_0 S_0^{\alpha}} > 0.$$

Then,

$$v_{\mathrm{d},l}(l) = C_1 S(l)^{\alpha - 1}.$$

Since the cross-sectional area satisfies

$$0 < S_{\min} \leq S(l) \leq S_{\max} < \infty,$$

and $0 < \alpha \leq 1$, it follows that $\alpha - 1 \leq 0$. Therefore,

$$v_{\mathrm{d},l}(l) = C_1 S(l)^{\alpha - 1} \geq C_1 S_{\max}^{\alpha - 1} > 0.$$

Hence, in this case $v_{\mathrm{d},l}$ does not decrease as S decreases, and it has the lower bound $C_1 S_{\max}^{\alpha - 1}$.

Next, it is proved that the desired density does not exceed the maximum density when

$$\alpha \leq \frac{\ln \left(\rho_{\max} / (w_{\max} \rho_0) \right)}{\ln \left(S_0 / S_{\min} \right)}. \tag{S20}$$

Since $S_0 > S_{\min}$ and $w_{\max} \rho_0 < \rho_{\max}$, both logarithms in Eq. (S20) are positive, so the admissible range of α is non-empty. Based on Eq. (S20), there is

$$-\alpha \ln \left(\frac{S_{\min}}{S_0} \right) \leq \ln \left(\frac{\rho_{\max}}{w_{\max} \rho_0} \right). \tag{S21}$$

Taking the exponential of both sides of Eq. (S21), there is

$$\rho_0 \left(\frac{S_{\min}}{S_0} \right)^{-\alpha} \leq \frac{\rho_{\max}}{w_{\max}}. \tag{S22}$$

Since $-\alpha < 0$ and $S(l) \geq S_{\min}$, it follows that $(S(l)/S_0)^{-\alpha} \leq (S_{\min}/S_0)^{-\alpha}$. According to Eq. (14) of the manuscript, $w(\boldsymbol{\zeta}) \leq w_{\max} = 1 + \mu_w$ on the unit cross section. Therefore, according to Eq. (13) of the manuscript and Eq. (S22),

$$\rho_{\mathrm{g}}(l, d_n, d_b) = \rho_{\mathrm{g},l}(l) w(\boldsymbol{\zeta}) \leq w_{\max} \rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha} \leq w_{\max} \rho_0 \left(\frac{S_{\min}}{S_0} \right)^{-\alpha} \leq \rho_{\max}.$$

Hence, the desired density satisfies the density constraint in Eq. (35) of the manuscript and is compatible with this constraint. This completes the proof.

Supplementary Appendix E. Mathematical Representation of VFF Planning. This appendix gives the parameterization and the discretized form of the optimization problem in Eq. (35) of the manuscript. The symbols are those defined in Section III of the manuscript.

1) Decision variables. In Eq. (35) of the manuscript, the decision variables are the longitudinal factor $\rho_{\mathrm{d},l}(l)$ of the planned density in Eq. (11) of the manuscript and the flux Q_0. The interval $[0, L]$ is divided into m_1 segments $[l_{k-1}, l_k]$, $k = 1, 2, \cdots, m_1$, with $l_0 = 0$, $l_{m_1} = L$, and length $h_k = l_k - l_{k-1}$. On the k-th segment, $\rho_{\mathrm{d},l}$ is an n-th order polynomial of the normalized arc length $s = (l - l_{k-1})/h_k \in [0, 1]$, so that $\mathrm{d}s/\mathrm{d}l = 1/h_k$. There is

$$\begin{aligned} \rho_{\mathrm{d},l}(l) &= \sum_{i=0}^{n} p_{i,k} s^i = \mathbf{s}_1^{\mathrm{T}} \mathbf{p}_k, \\ \rho'_{\mathrm{d},l}(l) &= \frac{\mathrm{d}\rho_{\mathrm{d},l}}{\mathrm{d}s}\frac{\mathrm{d}s}{\mathrm{d}l} = \frac{1}{h_k}\sum_{i=1}^{n} i p_{i,k} s^{i-1} = \frac{1}{h_k}\mathbf{s}_2^{\mathrm{T}} \mathbf{p}_k, \end{aligned} \tag{S23}$$

where

$$\mathbf{p}_k = \begin{bmatrix} p_{0,k} & p_{1,k} & \cdots & p_{n,k} \end{bmatrix}^{\mathrm{T}}, \quad \mathbf{s}_1 = \begin{bmatrix} 1 & s & s^2 & \cdots & s^n \end{bmatrix}^{\mathrm{T}}, \quad \mathbf{s}_2 = \begin{bmatrix} 0 & 1 & 2s & \cdots & n s^{n-1} \end{bmatrix}^{\mathrm{T}}.$$

The optimization vector is

$$\mathbf{x} = \begin{bmatrix} \mathbf{p}_1^{\mathrm{T}} & \mathbf{p}_2^{\mathrm{T}} & \cdots & \mathbf{p}_{m_1}^{\mathrm{T}} & Q_0 \end{bmatrix}^{\mathrm{T}} \in \mathbb{R}^{m_1(n+1)+1}.$$

For l in the k-th segment, define the row vectors

$$\begin{aligned} \mathbf{a}_1(l) &= \begin{bmatrix} \mathbf{0}_{1\times(k-1)(n+1)} & \mathbf{s}_1^{\mathrm{T}} & \mathbf{0}_{1\times(m_1-k)(n+1)} & 0 \end{bmatrix}, \\ \mathbf{a}_2(l) &= \begin{bmatrix} \mathbf{0}_{1\times(k-1)(n+1)} & \mathbf{s}_2^{\mathrm{T}}/h_k & \mathbf{0}_{1\times(m_1-k)(n+1)} & 0 \end{bmatrix}. \end{aligned}$$

Then, according to Eq. (S23), $\rho_{\mathrm{d},l}(l) = \mathbf{a}_1(l)\mathbf{x}$ and $\rho'_{\mathrm{d},l}(l) = \mathbf{a}_2(l)\mathbf{x}$, both linear in $\mathbf{x}$.

The value and the slope of $\rho_{\mathrm{d},l}$ are required to be continuous at the interior knots l_k, $k = 1, 2, \cdots, m_1 - 1$. Since $s = 1$ at the end of the k-th segment and $s = 0$ at the start of the $(k+1)$-th segment, there is

$$\begin{aligned} \mathbf{s}_1^{\mathrm{T}}(1)\mathbf{p}_k - \mathbf{s}_1^{\mathrm{T}}(0)\mathbf{p}_{k+1} &= \sum_{i=0}^{n} p_{i,k} - p_{0,k+1} = 0, \\ \frac{1}{h_k}\mathbf{s}_2^{\mathrm{T}}(1)\mathbf{p}_k - \frac{1}{h_{k+1}}\mathbf{s}_2^{\mathrm{T}}(0)\mathbf{p}_{k+1} &= \frac{1}{h_k}\sum_{i=1}^{n} i p_{i,k} - \frac{1}{h_{k+1}} p_{1,k+1} = 0. \end{aligned} \tag{S24}$$

Stacking Eq. (S24) for $k = 1, 2, \cdots, m_1 - 1$ gives the linear equality constraint

$$\mathbf{C}\mathbf{x} = \mathbf{0}_{2(m_1-1)},$$

where $\mathbf{C} \in \mathbb{R}^{2(m_1-1)\times(m_1(n+1)+1)}$ is

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \\ \vdots \\ \mathbf{C}_{m_1-1} \end{bmatrix}, \qquad \mathbf{C}_k = \begin{bmatrix} \mathbf{0}_{1\times(k-1)(n+1)} & \mathbf{s}_1^{\mathrm{T}}(1) & -\mathbf{s}_1^{\mathrm{T}}(0) & \mathbf{0}_{1\times(m_1-k-1)(n+1)} & 0 \\ \mathbf{0}_{1\times(k-1)(n+1)} & \frac{1}{h_k}\mathbf{s}_2^{\mathrm{T}}(1) & -\frac{1}{h_{k+1}}\mathbf{s}_2^{\mathrm{T}}(0) & \mathbf{0}_{1\times(m_1-k-1)(n+1)} & 0 \end{bmatrix},$$

where $k = 1, 2, \cdots, m_1 - 1$.

2) Planned density and velocity in terms of $\mathbf{x}$. Write $\boldsymbol{\zeta} = [\zeta_n, \zeta_b]^{\mathrm{T}} = [d_n, d_b]^{\mathrm{T}}/\lambda(l)$. According to Eq. (11) of the manuscript, the planned density is

$$\rho_{\mathrm{d}}(l, d_n, d_b) = \mathbf{a}_1(l)\mathbf{x}\, w(\boldsymbol{\zeta}). \tag{S25}$$

Since w has unit mean over the cross section, the line density in Eq. (15) of the manuscript is $M(l) = \mathbf{a}_1(l)\mathbf{x}\, S(l)$. According to Eq. (31) of the manuscript, the longitudinal velocity is

$$v_{\mathrm{d},l}(l) = \frac{Q_0}{\mathbf{a}_1(l)\mathbf{x}\, S(l)}. \tag{S26}$$

Taking the logarithm of Eq. (S26) and differentiating with respect to l, there is

$$\frac{v'_{\mathrm{d},l}(l)}{v_{\mathrm{d},l}(l)} = -\frac{S'(l)}{S(l)} - \frac{\mathbf{a}_2(l)\mathbf{x}}{\mathbf{a}_1(l)\mathbf{x}}.$$

Thus, there is

$$v'_{\mathrm{d},l}(l) = -v_{\mathrm{d},l}(l)\left(\frac{S'(l)}{S(l)} + \frac{\mathbf{a}_2(l)\mathbf{x}}{\mathbf{a}_1(l)\mathbf{x}}\right), \tag{S27}$$

where $S = 2\lambda$, $S' = 2\lambda'$ in 2D and $S = \pi\lambda^2$, $S' = 2\pi\lambda\lambda'$ in 3D.

According to Eq. (34) of the manuscript, there is

$$\begin{aligned} v_{\mathrm{d},d_n} &= \dot{d}_n - \frac{\tau d_b v_{\mathrm{d},l}}{J} = \frac{v_{\mathrm{d},l}\left(\lambda'\zeta_n - \tau\lambda\zeta_b\right)}{J}, \\ v_{\mathrm{d},d_b} &= \dot{d}_b + \frac{\tau d_n v_{\mathrm{d},l}}{J} = \frac{v_{\mathrm{d},l}\left(\lambda'\zeta_b + \tau\lambda\zeta_n\right)}{J}. \end{aligned} \tag{S28}$$

In 2D, $\tau = 0$, $\zeta = d_n/\lambda$, and Eq. (S28) reduces to $v_{\mathrm{d},d_n} = v_{\mathrm{d},l}\lambda'\zeta/J$ with $J = 1 - \kappa\lambda\zeta$. According to Eq. (S28), the speed is

$$\begin{aligned} \|\mathbf{v}_{\mathrm{d}}\|^2 &= v_{\mathrm{d},l}^2 + v_{\mathrm{d},d_n}^2 + v_{\mathrm{d},d_b}^2 \\ &= v_{\mathrm{d},l}^2\left[1 + \frac{\left(\lambda'\zeta_n - \tau\lambda\zeta_b\right)^2 + \left(\lambda'\zeta_b + \tau\lambda\zeta_n\right)^2}{J^2}\right] \\ &= v_{\mathrm{d},l}^2\left[1 + \frac{\left(\lambda'^2 + \tau^2\lambda^2\right)\|\boldsymbol{\zeta}\|^2}{J^2}\right]. \end{aligned} \tag{S29}$$

Since $Q_0 = \begin{bmatrix}\mathbf{0}_{1\times m_1(n+1)} & 1\end{bmatrix}\mathbf{x}$, substituting Eq. (S26) into Eq. (S29), there is

$$\|\mathbf{v}_{\mathrm{d}}\|^2 = \frac{\left(\begin{bmatrix}\mathbf{0}_{1\times m_1(n+1)} & 1\end{bmatrix}\mathbf{x}\right)^2}{\left(\mathbf{a}_1(l)\mathbf{x}\right)^2 S^2(l)}\left[1 + \frac{\left(\lambda'^2 + \tau^2\lambda^2\right)\|\boldsymbol{\zeta}\|^2}{J^2}\right]. \tag{S30}$$

3) Material derivatives. According to Eq. (32) of the manuscript, $\dot{d}_n = \dot{l}\lambda'\zeta_n$ and $\dot{d}_b = \dot{l}\lambda'\zeta_b$. Since $\zeta_n = d_n/\lambda$ and $\zeta_b = d_b/\lambda$, there is

$$\frac{\mathrm{D}\zeta_n}{\mathrm{D}t} = \frac{\dot{d}_n}{\lambda} - \frac{d_n\lambda'\dot{l}}{\lambda^2} = \frac{\dot{l}\lambda'\zeta_n}{\lambda} - \frac{\dot{l}\lambda'\zeta_n}{\lambda} = 0, \qquad \frac{\mathrm{D}\zeta_b}{\mathrm{D}t} = \frac{\dot{d}_b}{\lambda} - \frac{d_b\lambda'\dot{l}}{\lambda^2} = \frac{\dot{l}\lambda'\zeta_b}{\lambda} - \frac{\dot{l}\lambda'\zeta_b}{\lambda} = 0,$$

Thus, $\boldsymbol{\zeta}$ is constant. Substituting $\dot{d}_n = \dot{l}\,\lambda' d_n/\lambda$ and $\dot{d}_b = \dot{l}\,\lambda' d_b/\lambda$ into Eq. (18) of the manuscript, there is

$$\begin{aligned} \frac{\mathrm{D}g}{\mathrm{D}t} &= \dot{l}\frac{\partial g}{\partial l} + \dot{d}_n\frac{\partial g}{\partial d_n} + \dot{d}_b\frac{\partial g}{\partial d_b} \\ &= \dot{l}\left[\frac{\partial g}{\partial l} + \frac{\lambda'}{\lambda}\left(d_n\frac{\partial g}{\partial d_n} + d_b\frac{\partial g}{\partial d_b}\right)\right]. \end{aligned} \tag{S31}$$

Applying Eq. (S31) to $v_{\mathrm{d},l}$, which depends only on l, gives

$$\frac{\mathrm{D}v_{\mathrm{d},l}}{\mathrm{D}t} = \dot{l}\,v'_{\mathrm{d},l}. \tag{S32}$$

According to Eq. (S28) and $\boldsymbol{\zeta} = [d_n, d_b]^{\mathrm{T}}/\lambda$, there is

$$v_{\mathrm{d},d_n} = \frac{v_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)}{\lambda J}, \qquad v_{\mathrm{d},d_b} = \frac{v_{\mathrm{d},l}\left(\lambda' d_b + \tau\lambda d_n\right)}{\lambda J}. \tag{S33}$$

Since $\partial J/\partial l = -\kappa' d_n$, $\partial J/\partial d_n = -\kappa$, and $J + \kappa d_n = 1$, differentiating v_{d,d_n} in Eq. (S33) gives

$$\begin{aligned} \frac{\partial v_{\mathrm{d},d_n}}{\partial l} &= \frac{v'_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)}{\lambda J} + v_{\mathrm{d},l}\left[\frac{\lambda'' d_n - (\tau\lambda)' d_b}{\lambda J} - \frac{\left(\lambda' d_n - \tau\lambda d_b\right)\left(\lambda' J - \lambda\kappa' d_n\right)}{\lambda^2 J^2}\right], \\ d_n\frac{\partial v_{\mathrm{d},d_n}}{\partial d_n} + d_b\frac{\partial v_{\mathrm{d},d_n}}{\partial d_b} &= v_{\mathrm{d},l}\left[\frac{\lambda' d_n - \tau\lambda d_b}{\lambda J} + \frac{\left(\lambda' d_n - \tau\lambda d_b\right)\kappa d_n}{\lambda J^2}\right] = \frac{v_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)}{\lambda J^2}. \end{aligned} \tag{S34}$$

Substituting Eq. (S34) into Eq. (S31), there is

$$\begin{aligned} \frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t} &= \dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right) + v_{\mathrm{d},l}\left(\lambda'' d_n - (\tau\lambda)' d_b\right)}{\lambda J} - \frac{v_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)\left(\lambda' J - \lambda\kappa' d_n\right)}{\lambda^2 J^2} + \frac{v_{\mathrm{d},l}\lambda'\left(\lambda' d_n - \tau\lambda d_b\right)}{\lambda^2 J^2}\right] \\ &= \dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right) + v_{\mathrm{d},l}\left(\lambda'' d_n - (\tau\lambda)' d_b\right)}{\lambda J} + \frac{v_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)\left(\lambda'(1-J) + \lambda\kappa' d_n\right)}{\lambda^2 J^2}\right] \\ &= \dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right) + v_{\mathrm{d},l}\left(\lambda'' d_n - (\tau\lambda)' d_b\right)}{\lambda J} + \frac{v_{\mathrm{d},l}\left(\lambda' d_n - \tau\lambda d_b\right)(\kappa\lambda)' d_n}{\lambda^2 J^2}\right], \end{aligned} \tag{S35}$$

where $1-J=\kappa d_n$ and $\lambda'\kappa+\lambda\kappa'=(\kappa\lambda)'$ are used in the last line. In the same way, differentiating v_{d,d_b} in Eq. (S33) gives

$$\frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t}=\dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda' d_b+\tau\lambda d_n\right)+v_{\mathrm{d},l}\left(\lambda'' d_b+(\tau\lambda)' d_n\right)}{\lambda J}+\frac{v_{\mathrm{d},l}\left(\lambda' d_b+\tau\lambda d_n\right)(\kappa\lambda)' d_n}{\lambda^2 J^2}\right]. \tag{S36}$$

Substituting $d_n=\lambda\zeta_n$ and $d_b=\lambda\zeta_b$ into Eqs. (S35) and (S36), there is

$$\begin{aligned}\frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t}&=\dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)+v_{\mathrm{d},l}\left(\lambda''\zeta_n-(\tau\lambda)'\zeta_b\right)}{J}+\frac{v_{\mathrm{d},l}\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)(\kappa\lambda)'\zeta_n}{J^2}\right],\\ \frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t}&=\dot{l}\left[\frac{v'_{\mathrm{d},l}\left(\lambda'\zeta_b+\tau\lambda\zeta_n\right)+v_{\mathrm{d},l}\left(\lambda''\zeta_b+(\tau\lambda)'\zeta_n\right)}{J}+\frac{v_{\mathrm{d},l}\left(\lambda'\zeta_b+\tau\lambda\zeta_n\right)(\kappa\lambda)'\zeta_n}{J^2}\right].\end{aligned} \tag{S37}$$

According to the *Frenet–Serret formulas* $\mathbf{t}'=\kappa\mathbf{n}$, $\mathbf{n}'=-\kappa\mathbf{t}+\tau\mathbf{b}$, and $\mathbf{b}'=-\tau\mathbf{n}$, there is $\mathrm{d}\mathbf{t}/\mathrm{d}t=\kappa\dot{l}\,\mathbf{n}$, $\mathrm{d}\mathbf{n}/\mathrm{d}t=-\kappa\dot{l}\,\mathbf{t}+\tau\dot{l}\,\mathbf{b}$, and $\mathrm{d}\mathbf{b}/\mathrm{d}t=-\tau\dot{l}\,\mathbf{n}$. Differentiating Eq. (12) of the manuscript by the product rule gives

$$\begin{aligned}\frac{\mathrm{D}\mathbf{v}_{\mathrm{d}}}{\mathrm{D}t}&=\frac{\mathrm{D}v_{\mathrm{d},l}}{\mathrm{D}t}\mathbf{t}+v_{\mathrm{d},l}\frac{\mathrm{d}\mathbf{t}}{\mathrm{d}t}+\frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t}\mathbf{n}+v_{\mathrm{d},d_n}\frac{\mathrm{d}\mathbf{n}}{\mathrm{d}t}+\frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t}\mathbf{b}+v_{\mathrm{d},d_b}\frac{\mathrm{d}\mathbf{b}}{\mathrm{d}t}\\ &=\Big(\frac{\mathrm{D}v_{\mathrm{d},l}}{\mathrm{D}t}-\kappa\dot{l}\,v_{\mathrm{d},d_n}\Big)\mathbf{t}+\Big(\frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t}+\kappa\dot{l}\,v_{\mathrm{d},l}-\tau\dot{l}\,v_{\mathrm{d},d_b}\Big)\mathbf{n}+\Big(\frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t}+\tau\dot{l}\,v_{\mathrm{d},d_n}\Big)\mathbf{b}.\end{aligned}$$

Since $\mathbf{t}$, $\mathbf{n}$, and $\mathbf{b}$ are orthonormal, the acceleration magnitude is

$$\left\|\frac{\mathrm{D}\mathbf{v}_{\mathrm{d}}}{\mathrm{D}t}\right\|^2=\left(\frac{\mathrm{D}v_{\mathrm{d},l}}{\mathrm{D}t}-\kappa\dot{l}\,v_{\mathrm{d},d_n}\right)^2+\left(\frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t}+\kappa\dot{l}\,v_{\mathrm{d},l}-\tau\dot{l}\,v_{\mathrm{d},d_b}\right)^2+\left(\frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t}+\tau\dot{l}\,v_{\mathrm{d},d_n}\right)^2. \tag{S38}$$

Substituting $\dot{l}=v_{\mathrm{d},l}/J$, Eqs. (S26), (S27), (S28), (S32), and (S37) into Eq. (S38), with $Q_0=\begin{bmatrix}\mathbf{0}_{1\times m_1(n+1)} & 1\end{bmatrix}\mathbf{x}$, there is

$$\begin{aligned}\left\|\frac{\mathrm{D}\mathbf{v}_{\mathrm{d}}}{\mathrm{D}t}\right\|^2=\frac{\left(\begin{bmatrix}\mathbf{0}_{1\times m_1(n+1)} & 1\end{bmatrix}\mathbf{x}\right)^4}{\left(\mathbf{a}_1(l)\mathbf{x}\right)^4 S^4 J^2}\Bigg\{&\left[\left(\frac{S'}{S}+\frac{\mathbf{a}_2(l)\mathbf{x}}{\mathbf{a}_1(l)\mathbf{x}}\right)+\frac{\kappa\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)}{J}\right]^2\\ &+\left[\kappa-\left(\frac{S'}{S}+\frac{\mathbf{a}_2(l)\mathbf{x}}{\mathbf{a}_1(l)\mathbf{x}}\right)\frac{\lambda'\zeta_n-\tau\lambda\zeta_b}{J}+\frac{\lambda''\zeta_n-(\tau\lambda)'\zeta_b-\tau\left(\lambda'\zeta_b+\tau\lambda\zeta_n\right)}{J}\right.\\ &\quad\left.+\frac{\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)(\kappa\lambda)'\zeta_n}{J^2}\right]^2\\ &+\left[-\left(\frac{S'}{S}+\frac{\mathbf{a}_2(l)\mathbf{x}}{\mathbf{a}_1(l)\mathbf{x}}\right)\frac{\lambda'\zeta_b+\tau\lambda\zeta_n}{J}+\frac{\lambda''\zeta_b+(\tau\lambda)'\zeta_n+\tau\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)}{J}\right.\\ &\quad\left.+\frac{\left(\lambda'\zeta_b+\tau\lambda\zeta_n\right)(\kappa\lambda)'\zeta_n}{J^2}\right]^2\Bigg\}.\end{aligned} \tag{S39}$$

Apart from $\mathbf{x}$, Eq. (S39) contains only the geometric quantities of the virtual tube and $\boldsymbol{\zeta}$.

For the density, according to Eqs. (S25) and (S31), there is

$$\begin{aligned}\frac{\mathrm{D}\rho_{\mathrm{d}}}{\mathrm{D}t}&=\dot{l}\left[\frac{\partial\rho_{\mathrm{d}}}{\partial l}+\frac{\lambda'}{\lambda}\left(d_n\frac{\partial\rho_{\mathrm{d}}}{\partial d_n}+d_b\frac{\partial\rho_{\mathrm{d}}}{\partial d_b}\right)\right]\\ &=\dot{l}\left[\rho'_{\mathrm{d},l}w+\rho_{\mathrm{d},l}\left(\frac{\partial w}{\partial l}+\frac{\lambda'}{\lambda}\left(d_n\frac{\partial w}{\partial d_n}+d_b\frac{\partial w}{\partial d_b}\right)\right)\right]\\ &=\dot{l}\left[\rho'_{\mathrm{d},l}w+\rho_{\mathrm{d},l}\left(\frac{4\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}-\frac{4\mu_w\|\boldsymbol{\zeta}\|^2\lambda'}{\lambda}\right)\right]\\ &=\dot{l}\,\rho'_{\mathrm{d},l}w,\end{aligned} \tag{S40}$$

where Eq. (S10) is used in the third line. Substituting Eq. (S26) into Eq. (S40), there is

$$\frac{\mathrm{D}\rho_{\mathrm{d}}}{\mathrm{D}t}=\dot{l}\,\rho'_{\mathrm{d},l}\,w(\boldsymbol{\zeta})=\frac{v_{\mathrm{d},l}\,\mathbf{a}_2(l)\mathbf{x}\,w(\boldsymbol{\zeta})}{J}=\frac{Q_0\,\mathbf{a}_2(l)\mathbf{x}\,w(\boldsymbol{\zeta})}{J\,S(l)\,\mathbf{a}_1(l)\mathbf{x}}. \tag{S41}$$

4) Discretized optimization problem. Let $0=l_1<l_2<\cdots<l_P=L$ be the arc-length grid of the virtual tube. Since $M(l)/S(l)=\mathbf{a}_1(l)\mathbf{x}$, the term ρ_{obj} in Eq. (16) of the manuscript is

$$\rho_{\mathrm{obj}}(l)=\mathbf{a}_1(l)\mathbf{x}-\rho_0\left(\frac{S(l)}{S_0}\right)^{-\alpha},$$

and the two integrals in the cost function of Eq. (35) of the manuscript can be expressed as

$$\begin{aligned}
&\sum_{i=1}^{P-1} \frac{l_{i+1}-l_i}{2} \left[\frac{\omega_t}{Q_0} \Big(\mathbf{a}_1(l_i)\mathbf{x}\, S(l_i) + \mathbf{a}_1(l_{i+1})\mathbf{x}\, S(l_{i+1}) \Big) + \omega_\rho \Big(\, |\rho_{\text{obj}}(l_i)| + |\rho_{\text{obj}}(l_{i+1})| \, \Big) \right] \\
&= \sum_{i=1}^{P-1} \frac{l_{i+1}-l_i}{2} \left[\frac{\omega_t}{\begin{bmatrix} \mathbf{0}_{1\times m_1(n+1)} & 1 \end{bmatrix} \mathbf{x}} \Big(\mathbf{a}_1(l_i)\mathbf{x}\, S(l_i) + \mathbf{a}_1(l_{i+1})\mathbf{x}\, S(l_{i+1}) \Big) \right. \\
&\quad \left. + \omega_\rho \left(\left| \mathbf{a}_1(l_i)\mathbf{x} - \rho_0 \left(\frac{S(l_i)}{S_0} \right)^{-\alpha} \right| + \left| \mathbf{a}_1(l_{i+1})\mathbf{x} - \rho_0 \left(\frac{S(l_{i+1})}{S_0} \right)^{-\alpha} \right| \right) \right].
\end{aligned}$$

The constraints are imposed at the collocation points $(l_i, \boldsymbol{\zeta}_j)$, $i \in \mathcal{I}$, $j = 1, 2, \cdots, K_\zeta$, where $\mathcal{I} \subset \{1, 2, \cdots, P\}$ has $|\mathcal{I}|$ elements and $\boldsymbol{\zeta}_j$ lies in the unit cross section, $\zeta_j \in [-1, 1]$ in 2D and $\|\boldsymbol{\zeta}_j\| \leq 1$ in 3D. According to Eq. (S25) and $w > 0$, the density bounds in Eq. (35) of the manuscript are linear in $\mathbf{x}$,

$$\varepsilon \leq \mathbf{a}_1(l_i)\mathbf{x}\, w(\boldsymbol{\zeta}_j) \leq \rho_{\max}, \tag{S42}$$

where ε is a small positive number. Let $\mathcal{I} = \{i_1, i_2, \cdots, i_{|\mathcal{I}|}\}$. Stacking Eq. (S42) over all collocation points gives $\mathbf{B}\mathbf{x} \leq \mathbf{b}$, where

$$\mathbf{B} = \begin{bmatrix} w(\boldsymbol{\zeta}_1)\mathbf{a}_1(l_{i_1}) \\ \vdots \\ w(\boldsymbol{\zeta}_{K_\zeta})\mathbf{a}_1(l_{i_1}) \\ \vdots \\ w(\boldsymbol{\zeta}_{K_\zeta})\mathbf{a}_1(l_{i_{|\mathcal{I}|}}) \\ -w(\boldsymbol{\zeta}_1)\mathbf{a}_1(l_{i_1}) \\ \vdots \\ -w(\boldsymbol{\zeta}_{K_\zeta})\mathbf{a}_1(l_{i_{|\mathcal{I}|}}) \end{bmatrix} \in \mathbb{R}^{2|\mathcal{I}|K_\zeta \times (m_1(n+1)+1)}, \qquad \mathbf{b} = \begin{bmatrix} \rho_{\max}\mathbf{1}_{|\mathcal{I}|K_\zeta} \\ -\varepsilon\mathbf{1}_{|\mathcal{I}|K_\zeta} \end{bmatrix},$$

and $\mathbf{1}_{|\mathcal{I}|K_\zeta}$ is the $|\mathcal{I}|K_\zeta$-dimensional column vector of ones.

According to Eqs. (S30), (S39), and (S41), the remaining constraints of Eq. (35) of the manuscript are written as

$$\begin{aligned}
f_{1,ij}(\mathbf{x}) &= \frac{\left(\begin{bmatrix} \mathbf{0}_{1\times m_1(n+1)} & 1 \end{bmatrix} \mathbf{x} \right)^2}{\left(\mathbf{a}_1(l_i)\mathbf{x} \right)^2 S^2} \left[1 + \frac{\left(\lambda'^2 + \tau^2\lambda^2 \right) \|\boldsymbol{\zeta}_j\|^2}{J^2} \right] - v_{\max}^2 \leq 0, \\
f_{2,ij}(\mathbf{x}) &= \frac{\left(\begin{bmatrix} \mathbf{0}_{1\times m_1(n+1)} & 1 \end{bmatrix} \mathbf{x} \right)^4}{\left(\mathbf{a}_1(l_i)\mathbf{x} \right)^4 S^4 J^2} \left\{ \left[\left(\frac{S'}{S} + \frac{\mathbf{a}_2(l_i)\mathbf{x}}{\mathbf{a}_1(l_i)\mathbf{x}} \right) + \frac{\kappa \left(\lambda'\zeta_n - \tau\lambda\zeta_b \right)}{J} \right]^2 \right. \\
&\quad + \left[\kappa - \left(\frac{S'}{S} + \frac{\mathbf{a}_2(l_i)\mathbf{x}}{\mathbf{a}_1(l_i)\mathbf{x}} \right) \frac{\lambda'\zeta_n - \tau\lambda\zeta_b}{J} + \frac{\lambda''\zeta_n - (\tau\lambda)'\zeta_b - \tau \left(\lambda'\zeta_b + \tau\lambda\zeta_n \right)}{J} \right. \\
&\quad \left. + \frac{\left(\lambda'\zeta_n - \tau\lambda\zeta_b \right) (\kappa\lambda)'\zeta_n}{J^2} \right]^2 \\
&\quad + \left[- \left(\frac{S'}{S} + \frac{\mathbf{a}_2(l_i)\mathbf{x}}{\mathbf{a}_1(l_i)\mathbf{x}} \right) \frac{\lambda'\zeta_b + \tau\lambda\zeta_n}{J} + \frac{\lambda''\zeta_b + (\tau\lambda)'\zeta_n + \tau \left(\lambda'\zeta_n - \tau\lambda\zeta_b \right)}{J} \right. \\
&\quad \left. \left. + \frac{\left(\lambda'\zeta_b + \tau\lambda\zeta_n \right) (\kappa\lambda)'\zeta_n}{J^2} \right]^2 \right\} - a_v^2 \leq 0, \\
f_{3,ij}(\mathbf{x}) &= \left(\frac{\left(\begin{bmatrix} \mathbf{0}_{1\times m_1(n+1)} & 1 \end{bmatrix} \mathbf{x} \right) \mathbf{a}_2(l_i)\mathbf{x}\, w(\boldsymbol{\zeta}_j)}{J\, S\, \mathbf{a}_1(l_i)\mathbf{x}} \right)^2 - a_\rho^2 \leq 0,
\end{aligned}$$

where λ, λ', λ'', κ, $(\kappa\lambda)'$, τ, $(\tau\lambda)'$, S, and S' are evaluated at l_i, $[\zeta_n, \zeta_b]^{\mathrm{T}} = \boldsymbol{\zeta}_j$, and $J = 1 - \kappa(l_i)\lambda(l_i)\zeta_n$. Consequently,

the problem in Eq. (35) of the manuscript is discretized as

$$\begin{aligned}
\min_{\mathbf{x}} \quad & \sum_{i=1}^{P-1} \frac{l_{i+1}-l_i}{2}\left[\frac{\omega_t}{\begin{bmatrix}\mathbf{0}_{1\times m_1(n+1)} & 1\end{bmatrix}\mathbf{x}}\Big(\mathbf{a}_1(l_i)\mathbf{x}\,S(l_i)+\mathbf{a}_1(l_{i+1})\mathbf{x}\,S(l_{i+1})\Big)\right. \\
& \left.+\omega_\rho\left(\left|\mathbf{a}_1(l_i)\mathbf{x}-\rho_0\left(\frac{S(l_i)}{S_0}\right)^{-\alpha}\right|+\left|\mathbf{a}_1(l_{i+1})\mathbf{x}-\rho_0\left(\frac{S(l_{i+1})}{S_0}\right)^{-\alpha}\right|\right)\right] \\
\text{s.t.} \quad & \mathbf{C}\mathbf{x}=\mathbf{0}_{2(m_1-1)}, \\
& \mathbf{B}\mathbf{x}\le\mathbf{b}, \\
& f_{k,ij}(\mathbf{x})\le 0, \quad k=1,2,3,\ i\in\mathcal{I},\ j=1,2,\cdots,K_\zeta, \\
& Q_0\ge\varepsilon.
\end{aligned}$$

Supplementary Appendix F. The Proof of *Proposition 3*. This appendix provides a detailed proof of *Proposition 3* in Section III-C of the manuscript.

The proposition is proved by constructing a specific solution. Let $v_0>0$ and $C>0$ be constants, and take

$$\rho_{\mathrm{d},l}(l)=\frac{C}{v_0 S(l)}, \qquad Q_0=C, \qquad \rho_{\mathrm{d}}(l,d_n,d_b)=\frac{C\,w(\boldsymbol{\zeta})}{v_0 S(l)}. \tag{S43}$$

Then $M(l)=\rho_{\mathrm{d},l}(l)S(l)=C/v_0$, and Eq. (31) of the manuscript gives

$$v_{\mathrm{d},l}(l)=\frac{Q_0}{M(l)}=v_0, \qquad v'_{\mathrm{d},l}(l)=0, \qquad \dot{l}=\frac{v_0}{J}. \tag{S44}$$

According to Eqs. (S28) and (S44), the transverse velocity components are

$$v_{\mathrm{d},d_n}=\frac{v_0\left(\lambda'\zeta_n-\tau\lambda\zeta_b\right)}{J}, \qquad v_{\mathrm{d},d_b}=\frac{v_0\left(\lambda'\zeta_b+\tau\lambda\zeta_n\right)}{J}. \tag{S45}$$

Hence, every velocity component equals v_0 multiplied by a function of $(l,\boldsymbol{\zeta})$ that is determined by the virtual tube and does not depend on v_0 or C.

First, bounds on the geometric quantities appearing in the velocity and acceleration expressions are established. In 2D, $\lambda=S/2$, so $\lambda'=S'/2$ and $\lambda''=S''/2$. In 3D, $\lambda=\sqrt{S/\pi}$, so

$$\lambda'=\frac{S'}{2\sqrt{\pi S}}, \qquad \lambda''=\frac{S''}{2\sqrt{\pi S}}-\frac{S'^2}{4\sqrt{\pi}\,S^{3/2}}.$$

Since $S_{\min}\le S\le S_{\max}$, $|S'|\le\Delta_S$, and $|S''|$ is bounded, λ, λ', and λ'' are bounded on $[0,L]$. Together with the bounded κ, κ', τ, and τ', the derivatives $(\kappa\lambda)'=\kappa'\lambda+\kappa\lambda'$ and $(\tau\lambda)'=\tau'\lambda+\tau\lambda'$ are bounded. Since the tube does not fold, $\kappa\ge 0$, and $|\zeta_n|\le 1$, there is

$$J=1-\kappa\lambda\zeta_n\ge 1-\kappa\lambda\ge\min_{l\in[0,L]}\big(1-\kappa(l)\lambda(l)\big)>0. \tag{S46}$$

Next, it is proved that the speed constraint can be satisfied by choosing $v_0>0$ sufficiently small. According to Eqs. (S29), (S44), (S46), and $\|\boldsymbol{\zeta}\|\le 1$, there is

$$\|\mathbf{v}_{\mathrm{d}}\|^2=v_0^2\left[1+\frac{\left(\lambda'^2+\tau^2\lambda^2\right)\|\boldsymbol{\zeta}\|^2}{J^2}\right]\le v_0^2\max_{l\in[0,L]}\left[1+\frac{\lambda'^2+\tau^2\lambda^2}{(1-\kappa\lambda)^2}\right].$$

The maximum is finite, so $\|\mathbf{v}_{\mathrm{d}}\|\le v_{\max}$ holds if

$$v_0\le v_{\max}\left(\max_{l\in[0,L]}\left[1+\frac{\lambda'^2+\tau^2\lambda^2}{(1-\kappa\lambda)^2}\right]\right)^{-1/2}. \tag{S47}$$

Then, it is proved that the full acceleration constraint can also be satisfied by choosing $v_0>0$ sufficiently small.

Substituting $\dot{l} = v_0/J$ into Eq. (S38) and factoring out v_0^2/J gives

$$\left\| \frac{\mathrm{D}\mathbf{v}_\mathrm{d}}{\mathrm{D}t} \right\| = \frac{v_0^2}{J} \left\| \begin{bmatrix} -\kappa \dfrac{v_{\mathrm{d},d_n}}{v_0} \\ \kappa + \dfrac{J}{v_0^2} \dfrac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t} - \tau \dfrac{v_{\mathrm{d},d_b}}{v_0} \\ \dfrac{J}{v_0^2} \dfrac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t} + \tau \dfrac{v_{\mathrm{d},d_n}}{v_0} \end{bmatrix} \right\|. \tag{S48}$$

Since $v_{\mathrm{d},l} = v_0$ is constant, $\mathrm{D}v_{\mathrm{d},l}/\mathrm{D}t = 0$ and $v'_{\mathrm{d},l} = 0$. Dividing Eq. (S45) by v_0, and substituting $v_{\mathrm{d},l} = v_0$, $v'_{\mathrm{d},l} = 0$, and $\dot{l} = v_0/J$ into Eq. (S37) and multiplying by J/v_0^2, there is

$$\begin{aligned} \frac{v_{\mathrm{d},d_n}}{v_0} &= \frac{\lambda'\zeta_n - \tau\lambda\zeta_b}{J}, \\ \frac{J}{v_0^2}\frac{\mathrm{D}v_{\mathrm{d},d_n}}{\mathrm{D}t} &= \frac{\lambda''\zeta_n - (\tau\lambda)'\zeta_b}{J} + \frac{(\lambda'\zeta_n - \tau\lambda\zeta_b)(\kappa\lambda)'\zeta_n}{J^2}, \\ \frac{v_{\mathrm{d},d_b}}{v_0} &= \frac{\lambda'\zeta_b + \tau\lambda\zeta_n}{J}, \\ \frac{J}{v_0^2}\frac{\mathrm{D}v_{\mathrm{d},d_b}}{\mathrm{D}t} &= \frac{\lambda''\zeta_b + (\tau\lambda)'\zeta_n}{J} + \frac{(\lambda'\zeta_b + \tau\lambda\zeta_n)(\kappa\lambda)'\zeta_n}{J^2}. \end{aligned}$$

After substitution of these explicit expressions, the right-hand side of Eq. (S48) divided by v_0^2 is built from λ, λ', λ'', κ, $(\kappa\lambda)'$, τ, $(\tau\lambda)'$, $\boldsymbol{\zeta}$, and $1/J$, which are bounded by the geometric assumptions, Eq. (S46), and $\|\boldsymbol{\zeta}\| \leq 1$. Moreover, it does not depend on v_0 or C. Hence, its maximum over $l \in [0, L]$ is finite, and $\|\mathrm{D}\mathbf{v}_\mathrm{d}/\mathrm{D}t\| \leq a_v$ holds if

$$v_0 \leq \left(\frac{a_v}{\max\limits_{l\in[0,L],\, \|\boldsymbol{\zeta}\|\leq 1} \left(\|\mathrm{D}\mathbf{v}_\mathrm{d}/\mathrm{D}t\|/v_0^2 \right)} \right)^{1/2}, \tag{S49}$$

Finally, for a fixed $v_0 > 0$ satisfying the speed and acceleration constraints, it is proved that the density and its material derivative satisfy their respective bounds when $C > 0$ is sufficiently small. According to Eq. (14) of the manuscript, $0 < w(\boldsymbol{\zeta}) \leq w_{\max}$ for $0 \leq \mu_w < \bar{\mu}$. Hence, according to Eq. (S43),

$$0 < \rho_\mathrm{d} = \frac{C\, w(\boldsymbol{\zeta})}{v_0 S(l)} \leq \frac{w_{\max} C}{v_0 S_{\min}},$$

and $\rho_\mathrm{d} \leq \rho_{\max}$ holds if

$$C \leq \frac{v_0 S_{\min} \rho_{\max}}{w_{\max}}. \tag{S50}$$

According to Eqs. (S40), (S44), and $\rho'_{\mathrm{d},l} = -CS'/(v_0 S^2)$ from Eq. (S43), there is

$$\frac{\mathrm{D}\rho_\mathrm{d}}{\mathrm{D}t} = \dot{l}\, \rho'_{\mathrm{d},l}\, w(\boldsymbol{\zeta}) = \frac{v_0}{J} \left(-\frac{CS'}{v_0 S^2} \right) w(\boldsymbol{\zeta}) = -\frac{C\, S'\, w(\boldsymbol{\zeta})}{J\, S^2}.$$

According to Eq. (S46), $|S'| \leq \Delta_S$, and $S \geq S_{\min}$,

$$\left| \frac{\mathrm{D}\rho_\mathrm{d}}{\mathrm{D}t} \right| \leq \frac{C\, \Delta_S\, w_{\max}}{S_{\min}^2 \min_{l\in[0,L]} \left(1 - \kappa\lambda\right)},$$

and $|\mathrm{D}\rho_\mathrm{d}/\mathrm{D}t| \leq a_\rho$ holds if

$$C \leq \frac{a_\rho S_{\min}^2 \min_{l\in[0,L]} \left(1 - \kappa\lambda\right)}{w_{\max} \Delta_S}. \tag{S51}$$

In summary, by first selecting $v_0 > 0$ subject to Eqs. (S47) and (S49), and then $C > 0$ subject to Eqs. (S50) and (S51), the VFF given by Eqs. (S43)–(S45) is of the form required in Eq. (35) of the manuscript and satisfies constraints. This proves that the feasible set is non-empty.